

Reductive groups: generalities

SGA 3, Exposé XIX

English draft

M. Demazure

About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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Exposé XIX. Reductive groups—generalities

by *M. Demazure*

⁽⁰⁾The remainder of this Seminar is devoted to the study of reductive groups. Its principal aim is the generalization of the classical results of Chevalley (*Bible* and *Tôhoku*)⁽¹⁾ to arbitrary base schemes, the two central results being the *uniqueness theorem* (Exp. XXIII, th. 4.1 and cor. 5.1 to 5.10) and the *existence theorem* (Exp. XXV, th. 1.1) for “pinned” reductive group schemes corresponding to prescribed “root data”. The proofs used are inspired by those of Chevalley, the technique of schemes allowing them to be made more effective.

The results of the first volume of the *Bible* (Exp. 1 to 13) will be used systematically. On the other hand, we shall prove directly over an arbitrary scheme the results of the second volume (in particular, the isomorphism theorem); knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In the proof of these two fundamental results, we shall use only the most elementary results of the theory of groups of multiplicative type, contained for the most part in Exp. VIII and IX; on the other hand, we shall make essential use of the results of Exposé XVIII.⁽²⁾ The reader especially interested in the existence and uniqueness theorems may, on a first reading, skip Exposés X to XVII.

1. Reminders on groups over an algebraically closed field

1.1. — *In this section, k will always denote an algebraically closed field.* As announced above, the only results of *Bible* used in what follows are in volume 1 (Exposés 1 to 13). All the results of *Bible* (loc. cit.) concerning *semisimple groups* are valid more generally for *reductive groups* (definition below), and their proof is identical, with the following harmless modifications:

- Exp. 9, § 4, definition 3: see 1.6.1 below.
- Exp. 12, § 4, theorem 2, e): delete “finite”.
- Exp. 13, § 3, theorem 2: replace “rank” by “semisimple rank”.
- Exp. 13, § 4, corollary 2 to theorem 3: replace “rank” by “reductive rank”.

1.2. — Let G be a smooth affine connected k -group. The *radical* of G (*Bible*, § 9.4, prop. 2)⁽³⁾ is the reduced subgroup associated with the identity component of the intersection of the Borel subgroups of G ; it is also the largest smooth connected solvable invariant subgroup of G ; we shall denote it by $\text{rad}(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G ; it is also the largest smooth connected unipotent invariant subgroup of G ; we shall denote it by $\text{rad}^u(G)$.

⁽⁰⁾**N.D.E.** Version of 13/10/2024

⁽¹⁾**N.D.E.** See the bibliographical references at the end of this Exposé. In particular, a new edition of the Chevalley Seminar 1956–58, cited [Bible], revised by P. Cartier, was published in 2005, cf. [Ch05].

⁽²⁾**N.D.E.** More precisely, proposition XVIII 2.3 (extension of a “generic homomorphism” between groups) is used in XXII 4.1.11 and then in Exp. XXIII (proof of the uniqueness theorem), and also in Exp. XXIV; theorem XVIII 3.7 (construction of a group from a group germ) is used only in Exp. XXV.

⁽³⁾**N.D.E.** The reference 9-06 (= Exp. 9, p. 6) has been replaced by § 9.4 (= Exp. 9, § 4), which applies equally to [Bible] and [Ch05]. When the numbering of [Ch05] differs from [Bible], as is the case in Exp. 6, both references will be indicated explicitly.

1.3. — Let G be a smooth affine connected k -group, and Q a torus of G . Then the centralizer $\text{Centr}_G(Q)$ of Q in G is a closed subgroup of G (Exp. VIII, 6.7), smooth (Exp. XI, 2.4) and connected (cf. *Bible*, § 6.6, th. 6 a) or [Ch05], § 6.7, th. 7 a)).

One has the fundamental relation:

$$(1.3.1) \quad \text{rad}^u(\text{Centr}_G(Q)) = \text{rad}^u(G) \cap \text{Centr}_G(Q).$$

⁽⁴⁾First, $U = \text{rad}^u(G) \cap \text{Centr}_G(Q)$ is a unipotent invariant subgroup of $\text{Centr}_G(Q)$; let us show that it is smooth and connected. If one makes Q act on $\text{rad}^u(G)$ by inner automorphisms, U is precisely the scheme of invariants of this action. Now this is smooth and connected, by lemma 1.4 below.

Consequently, U is a closed subgroup of $\text{rad}^u(\text{Centr}_G(Q))$. On the other hand, by *Bible*, Exp. 12, § 3, cor. to th. 1, one has $\text{rad}^u(\text{Centr}_G(Q))(k) \subset \text{rad}^u(G)(k)$. Equality (1.3.1) follows.

Let us point out a particular case of the preceding result: if G is a smooth affine connected k -group and T is a maximal torus of G ,

$$(1.3.2) \quad \text{Centr}_G(T) = T \cdot (\text{Centr}_G(T) \cap \text{rad}^u(G)).$$

Indeed (cf. *Bible*, § 6.6, th. 6 c) or [Ch05], § 6.7, th. 7 c)), $\text{Centr}_G(T)$ is a solvable group, hence the semidirect product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, § 6.5, th. 5 a) or [Ch05], § 6.6, th. 6 a)), the union of the $\text{Centr}_G(T)$, as T runs through the set of maximal tori of G , contains a dense open subset of G ; it therefore follows from (1.3.2):

Corollary 1.3.3. *If G is a smooth affine connected k -group, the union of the $T \cdot \text{rad}^u(G)$, where T runs through the set of maximal tori of G , contains a dense open subset of G .*

Notation 1.4.0. ⁽⁵⁾We shall systematically use the following notation: if S is a scheme and s a point of S , $\bar{s}$ will denote the spectrum of an algebraic closure $\kappa(\bar{s})$ of $\kappa(s)$.

Lemma 1.4. ⁽⁵⁾Let S be a scheme, Q an S -torus acting on a group S -scheme H , separated and smooth over S .

- (i) *The functor of invariants H^Q is representable by a closed subscheme of H , smooth over S .*
- (ii) *If in addition H is affine over S , and has connected fibers, then so does H^Q .*

First, by VIII 6.5 (d), ⁽⁶⁾ since H is separated over S and Q essentially free over S , H^Q is representable by a closed subscheme of H . In particular, if H is affine over S , then so is H^Q .

Consider now the semidirect product $G = H \cdot Q$; it is smooth and separated over S . Then the centralizer $\text{Centr}_G(Q)$ is representable by a closed subscheme of G , of finite presentation over G , by Exp. XI 6.11 (a), ⁽⁶⁾ and it is smooth over S by Exp. XI 2.4.

Since $\text{Centr}_G(Q) = H^Q \cdot Q$, it follows that H^Q is smooth over S . (Indeed, using the section $\sigma : H^Q \rightarrow \text{Centr}_G(Q)$ deduced from the unit section ε_Q of Q , one sees at once that H^Q is formally smooth over S ; since moreover ε_Q is of finite presentation, H^Q is locally of finite presentation over S .) This proves (i).

Finally, suppose that H is affine over S and has connected fibers. Then each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$ -group, hence, by the first assertion of 1.3, the same is true of

$$\text{Centr}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\text{Centr}_G(Q))_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}},$$

and this implies that $(H^Q)_{\bar{s}}$ is connected.

⁽⁴⁾**N.D.E.** The following has been expanded.

⁽⁵⁾**N.D.E.** These notations, which appeared in 2.3 of the original, have been inserted here; on the other hand, the statement and proof of 1.4 have been expanded.

⁽⁶⁾**N.D.E.** See also the additions made in VI_B, 6.2.4 (d) and 6.5.5 (a).

1.5. — Recall that the *reductive rank* of the smooth affine k -group G is the common dimension of the maximal tori of G . We shall denote it by $\mathrm{rgred}(G/k)$ or $\mathrm{rgred}(G)$. For $\mathrm{rgred}(G/k) = 0$, it is necessary and sufficient that G be unipotent (i.e. that $G = \mathrm{rad}^u(G)$), by *Bible*, § 6.4, cor. 1 to th. 4 or [Ch05], § 6.5, cor. 1 to th. 5.

If H is an invariant subgroup of the smooth affine k -group G , the quotient G/H is affine and smooth (Exp. VI_B, 11.17 and VI_A, 3.3.2 (iii)). Moreover (*Bible*, § 7.3, th. 3, a) and c)), one has

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/H) + \mathrm{rgred}(H).$$

Definition 1.6.1. ⁽⁷⁾One says that the k -group G is *reductive* if it is smooth, affine and connected, and if $\mathrm{rad}(G)$ is a torus, that is, if $\mathrm{rad}^u(G) = \{e\}$.

Lemma 1.6.2. *Let G be a reductive k -group.*

- (i) *If Q is a torus of G , then $\mathrm{Centr}_G(Q)$ is reductive.*
- (ii) *In particular, if T is a maximal torus of G , then $\mathrm{Centr}_G(T) = T$.*
- (iii) *The center of G is contained in every maximal torus, hence is diagonalizable.*
- (iv) *The radical of G is the (unique) maximal torus of $\mathrm{Centr}(G)$.*

Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $\mathrm{Centr}(G)$ (i.e. the identity component $\mathrm{Centr}(G)^0$) is a smooth solvable connected subgroup, invariant in G , hence contained in $\mathrm{rad}(G)$; conversely, $\mathrm{rad}(G)$ is a torus invariant in G , hence central (*Bible*, § 4.3, cor. to prop. 2), whence (iv).

Remark 1.6.3. If G is reductive and $\dim(G) \neq \mathrm{rgred}(G)$, then $\dim(G) - \mathrm{rgred}(G) \geq 2$. Indeed, this difference is always even (cf. 1.10 below).

1.7. — Let G be a smooth affine connected k -group and H a smooth connected invariant subgroup. Then

$$(1.7.1) \quad \mathrm{rad}(H) = (\mathrm{rad}(G) \cap H)_{\mathrm{red}}^0 \quad \text{and} \quad \mathrm{rad}^u(H) = (\mathrm{rad}^u(G) \cap H)_{\mathrm{red}}^0,$$

as one sees at once. In particular, if G is reductive, so is H .

Let $f : G \rightarrow G'$ be a surjective morphism of smooth affine connected k -groups. Then

$$(1.7.2) \quad f(\mathrm{rad}^u(G)) = \mathrm{rad}^u(G').$$

In particular, if G is reductive, so is G' .

Let us prove (1.7.2). First, f sends $\mathrm{rad}^u(G)$ into $\mathrm{rad}^u(G')$. Introducing

$$H = (f^{-1}(\mathrm{rad}^u(G'))_{\mathrm{red}})^0,$$

which contains $\mathrm{rad}^u(G)$, one has $\mathrm{rad}^u(H) = \mathrm{rad}^u(G)$ and one is reduced to the case where $G = H$, i.e. where G' is unipotent. Since the union of the $T \cdot \mathrm{rad}^u(G)$ (T running through the set of maximal tori of G) is dense in G , the union of the $f(T)f(\mathrm{rad}^u(G))$ is dense in G' ; but $f(T)$ consists of semisimple elements, hence $f(T) = \{e\}$, since G' is unipotent; this shows that $f(\mathrm{rad}^u(G))$ is dense in G' . Therefore, by *Bible*, § 5.4, lemma 4 or [Ch05], § 6.1, lemma 1,⁽⁸⁾ $f(\mathrm{rad}^u(G))$ is an open subgroup of G' ; since the latter is connected, it follows that $f(\mathrm{rad}^u(G)) = G'$.

(N. B.: under the same hypotheses one can prove that $f(\mathrm{rad}(G)) = \mathrm{rad}(G')$.)

1.8. — One says that the k -group G is *semisimple* if it is smooth, affine and connected and if $\mathrm{rad}(G) = \{e\}$, that is, if G is reductive and $\mathrm{Centr}(G)$ finite. If G is any smooth affine connected k -group, then $G/\mathrm{rad}(G)$ is semisimple (*Bible*, § 9.4, prop. 2), and $G/\mathrm{rad}^u(G)$ is reductive. One calls *semisimple rank* of G , and denotes by $\mathrm{rgss}(G/k)$ or $\mathrm{rgss}(G)$, the reductive rank of $G/\mathrm{rad}(G)$.

⁽⁷⁾**N.D.E.** Paragraph 1.6 has been changed into paragraphs 1.6.1 to 1.6.3. Note that in this Seminar every reductive (resp. semisimple, cf. 1.8) k -group is, by definition, connected.

⁽⁸⁾**N.D.E.** The following has been expanded.

If G is reductive, one therefore has

$$\mathrm{rgred}(G) = \mathrm{rgss}(G) + \dim(\mathrm{rad}(G)).$$

If G is a smooth affine connected k -group and Q is a central subtorus of G , then G/Q is semisimple if and only if G is reductive and $Q = \mathrm{rad}(G)$. Indeed, if G/Q is semisimple, $Q \supset \mathrm{rad}(G)$, hence $\mathrm{rad}(G)$ is a torus, hence G is reductive, hence $\mathrm{rad}(G)$ is the maximal torus of $\underline{\mathrm{Centr}}(G)$, hence $\mathrm{rad}(G) = Q$. If G is reductive and Q is a central subgroup, then (G/Q is reductive and) $\mathrm{rgss}(G) = \mathrm{rgss}(G/Q)$.

1.9. — If K is an algebraically closed extension of k and G is a smooth affine connected k -group, G is reductive (resp. semisimple) if and only if G_K is, and one has

$$\mathrm{rgred}(G/k) = \mathrm{rgred}(G_K/K), \quad \mathrm{rgss}(G/k) = \mathrm{rgss}(G_K/K).$$

1.10. — Let G be a smooth connected k -group and T a torus of G .⁽⁹⁾ One denotes by $\mathfrak{g}$ the Lie k -algebra of G , i.e. $\mathfrak{g} = \mathrm{Lie}(G) = \mathrm{Lie}(G/k)(k)$; similarly, one puts $\mathfrak{t} = \mathrm{Lie}(T)$. Then $\mathfrak{g}$ decomposes under the action of T (by $T \rightarrow G$ and the adjoint action of G) into

$$(1.10.1) \quad \mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha,$$

where the $\alpha \in R$ are nontrivial characters of T and the $\mathfrak{g}^\alpha$ are $\neq 0$. The characters $\alpha \in R$ are called the *roots* of G with respect to T . By Exp. II, 5.2.3, (i), one has

$$(1.10.2) \quad \mathfrak{g}^0 = \mathrm{Lie}(\underline{\mathrm{Centr}}_G(T)).$$

In particular,⁽¹⁰⁾ since $\underline{\mathrm{Centr}}_G(T)$ is connected (1.3 when G is affine, and Exp. XII 6.6 b) in the general case), T is its own centralizer if and only if $\mathfrak{g}^0 = \mathrm{Lie}(T)$.

This condition implies that T is maximal and that $\underline{\mathrm{Centr}}(G) \subset T$, hence by Exp. XII 8.8 d) one has:

$$(1.10.3) \quad \underline{\mathrm{Centr}}(G) = \mathrm{Ker}(T \rightarrow \mathrm{GL}(\mathfrak{g})) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha);$$

moreover, $\underline{\mathrm{Centr}}(G)$ is then affine, hence $G \rightarrow G/\underline{\mathrm{Centr}}(G)$ affine; since the latter group is affine (Exp. XII 6.1),⁽¹¹⁾ so is G .

When G is reductive and T maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, § 12.2, def. 1; the latter are indeed roots in the sense of this paragraph (*Bible*, § 13.2, th. 1, c)) and there are $\dim(G) - \dim(T)$ of them (*Bible*, § 13.4, cor. 2 to th. 3). Moreover, if α is a root, so is $-\alpha$ (*Bible*, § 12.2, cor. to prop. 1). (As usual, one writes the group structure of $\mathrm{Hom}_{k\text{-gr.}}(T, \mathbf{G}_{m,k})$ indifferently additively or multiplicatively.) It follows that, for G reductive,

$$(1.10.4) \quad \dim(G) - \mathrm{rgred}(G) = \mathrm{Card}(R)$$

is always even.

The semisimple rank of the reductive group G is the rank of the subset R of the $\mathbf{Q}$ -vector space $\mathrm{Hom}_{k\text{-gr.}}(T, \mathbf{G}_{m,k}) \otimes \mathbf{Q}$ (*Bible*, § 13.3, th. 2).

Lemma 1.11. *Let k be an algebraically closed field, G a smooth affine connected k -group, T a torus of G , $W(T) = \underline{\mathrm{Norm}}_G(T)/\underline{\mathrm{Centr}}_G(T)$ the Weyl group of G with respect to T . The following conditions are equivalent:*

- (i) G is reductive, T is maximal, $\mathrm{rgss}(G) = 1$.
- (ii) G is reductive, T is maximal, $G \neq T$; there exists a subtorus Q of T , of codimension 1 in T , central in G .
- (iii) G is not solvable and $\dim(G) - \dim(T) \leq 2$.

⁽⁹⁾**N.D.E.** The following sentence has been added.

⁽¹⁰⁾**N.D.E.** The following has been expanded.

⁽¹¹⁾**N.D.E.** See also the addition VI_B, 6.2.6.

- (iv) $W(T) \neq \{e\}$ and $\dim(G) - \dim(T) \leq 2$.
- (v) $W(T) = (\mathbf{Z}/2\mathbf{Z})_k$ and $\dim(G) - \dim(T) = 2$.

Moreover, under these conditions, there are exactly two roots of G with respect to T ; they are opposite. Under the conditions of (ii), $Q = \text{rad}(G)$.

One clearly has (v) $\Rightarrow$ (iv). One has (iv) $\Rightarrow$ (iii) by *Bible*, § 6.1, cor. 3 to th. 1 or [Ch05], § 6.2, cor. 3 to th. 2. Let us prove (iii) $\Rightarrow$ (ii): let U be the unipotent radical of G ; one knows that G/U is reductive and is not a torus (otherwise G would be solvable); hence, by (1.10.4), one has

$$\dim(G/U) - \text{rgred}(G/U) = \text{Card}(R) \geq 2;$$

but

$$\text{rgred}(G) = \text{rgred}(G/U) + \text{rgred}(U) = \text{rgred}(G/U),$$

whence

$$\begin{aligned} \dim(G) - \dim(U) - \text{rgred}(G) &= \dim(G/U) - \text{rgred}(G/U) \\ &\geq 2 \geq \dim(G) - \dim(T) \geq \dim(G) - \text{rgred}(G). \end{aligned}$$

This gives $\dim(U) = 0$, hence G is reductive, $\text{rgred}(G) = \dim(T)$, hence T is maximal, $\dim(G) - \dim(T) = 2$. By *Bible*, § 10.4, prop. 8, there exists a singular torus Q of codimension 1 in T ; then $\text{Centr}_G(Q)$ is reductive (1.6.2 (i)), not solvable (definition of a singular torus), hence $\dim(\text{Centr}_G(Q)) - \text{rgred}(G) \geq 2$, which proves that $G = \text{Centr}_G(Q)$ and Q is central in G .

Let us prove (ii) $\Rightarrow$ (i). One knows that G/Q is reductive (1.7) and that $\text{rgred}(G/Q) = 1$, hence $\text{rgss}(G/Q) = 0$ or 1. The first case is impossible, for it would imply $\text{rgss}(G) = 0$, hence $G = T$; one therefore has $\text{rgred}(G/Q) = \text{rgss}(G/Q) = 1$, which proves that G/Q is semisimple; hence Q is the radical of G and $\text{rgss}(G) = 1$. Let us finally prove (i) $\Rightarrow$ (v). If Q is the radical of G , one has $\dim(T) - \dim(Q) = 1$ and Q is central in G , hence $G = \text{Centr}_G(Q)$, which proves that Q is a singular torus; by *Bible*, § 11.3, th. 2, one has $W_G(T) = (\mathbf{Z}/2\mathbf{Z})_k$; by *Bible*, § 12.1, lemma 1, one has $\dim(G) - \dim(T) = 2$. There are therefore at most two roots of G with respect to T , whereas there are at least two, opposite (1.10).

Proposition 1.12. *Let k be an algebraically closed field, G a smooth connected k -group, T a torus of G , R the set of roots of G with respect to T , and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha, \quad \text{with } \mathfrak{g}^\alpha \neq 0,$$

the decomposition of $\mathfrak{g}$ under $\text{Ad}(T)$. For each $\alpha \in R$, let T_α be the maximal torus of $\text{Ker}(\alpha)$,⁽¹²⁾ and $Z_\alpha = \text{Centr}_G(T_\alpha)$. The following conditions are equivalent:

- (i) G is affine, reductive; T is maximal.
- (ii) $\mathfrak{g}^0 = \mathfrak{t}$, each Z_α ($\alpha \in R$) is reductive.
- (iii) $\mathfrak{g}^0 = \mathfrak{t}$, each $\mathfrak{g}^\alpha$ ($\alpha \in R$) has dimension 1; and if $\alpha, \beta \in R$ and $q \in \mathbf{Q}$ are such that $\beta = q\alpha$, then $q = \pm 1$; for each $\alpha \in R$, there exists a $w_\alpha \in G(k)$ which normalizes T , centralizes T_α , but does not centralize T .

Moreover, under these conditions, each Z_α has semisimple rank 1 and one has $\text{Lie}(Z_\alpha) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$.

If G is affine and reductive and T is maximal, each Z_α is affine and reductive (1.6.2 (i)), with maximal torus T ; moreover, T is its own centralizer (1.6.2 (ii)), hence $\mathfrak{g}^0 = \mathfrak{t}$, which proves (i) $\Rightarrow$ (ii).

On the other hand, $\mathfrak{g}^0 = \mathfrak{t}$ implies that T is maximal and G affine (cf. 1.10), hence each Z_α is affine, smooth and connected, by 1.3. In any case, by Exp. II, 5.2.2, one has

$$\text{Lie}(Z_\alpha) = \mathfrak{g}^{T_\alpha} = \mathfrak{g}^0 \oplus \coprod_{\beta \in R \cap \mathbf{Q}\alpha} \mathfrak{g}^\beta.$$

⁽¹²⁾**N.D.E.** I.e. the connected component of $\text{Ker}(\alpha)$.

One therefore has

$$\mathrm{Lie}(Z_\alpha) \supset \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

which implies $Z_\alpha \neq T$. Since T_α is a subtorus of codimension 1 in T , central in Z_α , one obtains by 1.11, applied to Z_α , the equivalence (ii) $\Leftrightarrow$ (iii) and the supplementary assertions.

It remains to prove (ii) $\Rightarrow$ (i); one already knows that (ii) implies that T is maximal and G affine. Let U be its unipotent radical; it is invariant in G , its Lie algebra is invariant under $\mathrm{Ad}(T)$. One therefore has

$$\mathrm{Lie}(U) = (\mathrm{Lie}(U) \cap \mathfrak{g}^0) + \coprod_{\alpha \in R} (\mathrm{Lie}(U) \cap \mathfrak{g}^\alpha).$$

By (1.3.1), one has

$$\begin{aligned} U \cap T &= U \cap \mathrm{Centr}_G(T) = \mathrm{rad}^u(\mathrm{Centr}_G(T)) = \mathrm{rad}^u(T) = \{e\}, \\ U \cap Z_\alpha &= U \cap \mathrm{Centr}_G(T_\alpha) = \mathrm{rad}^u(\mathrm{Centr}_G(T_\alpha)) = \mathrm{rad}^u(Z_\alpha) = \{e\}. \end{aligned}$$

One therefore has⁽¹³⁾

$$\begin{aligned} \mathrm{Lie}(U) \cap \mathfrak{g}^0 &= \mathrm{Lie}(U) \cap \mathfrak{t} = \mathrm{Lie}(U \cap T) = 0, \\ \mathrm{Lie}(U) \cap \mathfrak{g}^\alpha &\subset \mathrm{Lie}(U) \cap \mathrm{Lie}(Z_\alpha) = \mathrm{Lie}(U \cap Z_\alpha) = 0; \end{aligned}$$

whence $\mathrm{Lie}(U) = 0$ and $U = \{e\}$, i.e. G is reductive.

Corollary 1.13. *Let k be an algebraically closed field, G a smooth connected k -group, T a torus of G , R the set of roots of G with respect to T , and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$$

as above. For each $\alpha \in R$, let T_α be the maximal torus of $\mathrm{Ker}(\alpha)$ and $Z_\alpha = \mathrm{Centr}_G(T_\alpha)$. The following conditions are equivalent:

- (i) *G is affine, semisimple; T is maximal.*
- (ii) *$\mathfrak{g}^0 = \mathfrak{t}$, each Z_α is reductive and $\bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$ is finite.*

⁽¹⁴⁾This follows from the equalities $\mathrm{rad}(G) = \mathrm{Centr}(G)^0$ and $\mathrm{Centr}(G) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha)$, established in 1.6.2 (iv) and (1.10.3).

2. Reductive group schemes. Definitions and first properties

Scholium 2.1. *If G is a group scheme over S , the following properties are equivalent:*

- (i) *G is affine and smooth over S , and has connected fibers.*
- (ii) *G is affine and flat over S , of finite presentation over S , and has geometrically integral fibers.*

These properties are stable under base change and local for (fpqc).

⁽¹⁵⁾Indeed, suppose (i) holds. Since G is affine and smooth over S , it is of finite presentation over S ; and since its fibers are smooth and connected, they are geometrically integral, by VI_A, 2.4.

Conversely, if (ii) holds, the fibers of G are connected and geometrically reduced, hence smooth (VI_A, 1.3.1); then G is smooth over S , by EGA IV₄, 17.5.2.

Of course, these properties are stable under base change: cf. EGA II, 1.5.1 for “affine”, IV₁, 1.6.2 (iii) for “of finite presentation”, IV₂, 2.1.4 for “flat”, and IV₂, 4.6.5 (i) for “with geometrically integral fibers”.

⁽¹³⁾**N.D.E.** Note that if U, L are two subgroup schemes of G , one has $\mathrm{Lie}(U) \cap \mathrm{Lie}(L) = \mathrm{Lie}(U \cap L)$.

⁽¹⁴⁾**N.D.E.** The following sentence has been added.

⁽¹⁵⁾**N.D.E.** The following has been added.

On the other hand, these properties are clearly local for the Zariski topology, so it suffices to check that if $S' \rightarrow S$ is a faithfully flat quasi-compact morphism and $G_{S'} \rightarrow S'$ has the indicated properties, the same is true of $G \rightarrow S$. This follows from EGA IV₂, 2.5.1 for “flat”, 2.7.1 (vi) and (xiii) for “of finite presentation” and “affine”, and 4.6.5 (i) for “with geometrically integral fibers” (and also EGA IV₄, 17.7.3 (ii) for “smooth”).

2.2. — Let G be a group S -scheme satisfying the preceding conditions, and Q a torus (cf. IX, Def. 1.3) of G .⁽¹⁶⁾ Then, by XI, 6.11 a) and XI, 2.4, $\text{Centr}_G(Q)$ is representable by a closed subgroup scheme of G (hence affine over S), of finite presentation and smooth over S ; moreover, since each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$ -group, by the first assertion of 1.3 the same is true of

$$\text{Centr}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\text{Centr}_G(Q))_{\bar{s}}.$$

Lemma 2.3. *Let S be a scheme, G a group S -scheme smooth and affine over S , with connected fibers, and T a torus of G . The set of $s \in S$ such that $G_{\bar{s}}$ is a reductive $\bar{s}$ -group, of semisimple rank 1 and with maximal torus $T_{\bar{s}}$, is an open subset U of S .*

Since G and T are flat over S , the function

$$s \mapsto \dim(G_{\bar{s}}/\bar{s}) - \dim(T_{\bar{s}}/\bar{s})$$

is locally constant; let U_1 be the open subset of points of S where its value is 2.

⁽¹⁷⁾By 6.3, the Weyl group

$$W_G(T) = \text{Norm}_G(T)/\text{Centr}_G(T)$$

is representable by a group S -scheme *étale and separated* over S , and the function

$$s \mapsto \text{number of points of } W_G(T)_{\bar{s}}$$

is lower semicontinuous. Let U_2 be the set of points of S where this function is > 1 ; it is open. By 1.11, the set of s such that $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is $U = U_1 \cap U_2$; moreover, for every $s \in U$, $W_G(T)_{\bar{s}}$ has exactly two points.

Consequently (cf. SGA 1, I 10.9 and EGA IV₃, 15.5.1 and IV₄, 18.12.4), $W_G(T)_U$ is *étale and finite* over U .

Remark 2.4. The group $W_G(T)_U$ is *isomorphic* to $(\mathbf{Z}/2\mathbf{Z})_U$. Indeed, by what precedes, it is *étale and finite* over U ; since the functor of automorphisms of $(\mathbf{Z}/2\mathbf{Z})_U$ is the unit group, it suffices to check the assertion locally, and it is immediate if the algebra $\mathcal{A}$ defining $W_G(T)_U$ is a free $\mathcal{O}_U$ -module.⁽¹⁸⁾

Notations and reminders 2.5.0. ⁽¹⁹⁾Let S be a scheme, G a group S -scheme, $\varepsilon : S \rightarrow G$ the unit section of G . We saw in II, § 4.11 that the functor $\text{Lie}(G/S)$ is representable by the vector bundle $V(\omega_{G/S}^1)$ (where $\omega_{G/S}^1 = \varepsilon^*(\Omega_{G/S}^1)$), and one denotes by

$$\mathfrak{g} = \text{Lie}(G/S) = (\omega_{G/S}^1)^\vee$$

the sheaf of sections of this vector bundle. Suppose in addition that G is *smooth* over S ; then $\omega_{G/S}^1$ and hence $\text{Lie}(G/S)$ are locally free $\mathcal{O}_S$ -modules of finite type, and one has (cf. I

⁽¹⁶⁾**N.D.E.** The following has been expanded.

⁽¹⁷⁾**N.D.E.** The following has been expanded. Note, on the other hand, that section 6 is independent of the rest of this Exposé.

⁽¹⁸⁾**N.D.E.** The hypothesis implies that $\mathcal{A}$ is locally free of rank 2, and since the augmentation ideal $\mathcal{I}$ is a direct summand of $\mathcal{A}$, replacing U by a sufficiently small affine open subset $\text{Spec}(R)$, one may suppose that $I = \Gamma(U, \mathcal{I})$ is a free R -module of rank 1. If x is a generator of I , one then has $x^2 = ax$ for some $a \in R$, and the hypothesis that $A = \Gamma(U, \mathcal{A})$ is *étale* implies that a is invertible in R , and one then easily sees that A is the affine algebra of the constant R -group $\mathbf{Z}/2\mathbf{Z}$ (compare with the first lines of [TO70]).

⁽¹⁹⁾**N.D.E.** This paragraph of notations and reminders has been added.

4.6.5.1):

$$\underline{\mathrm{Lie}}(G/S) = \mathrm{W}(\mathcal{L}ie(G/S)),$$

i.e. for every S -scheme S' ,

$$\underline{\mathrm{Lie}}(G/S)(S') = \Gamma(S', \mathcal{L}ie(G/S) \otimes \mathcal{O}_{S'}).$$

By II 4.1.1.1, the adjoint action of G equips $\underline{\mathrm{Lie}}(G/S) = \mathrm{W}(\mathcal{L}ie(G/S))$ with a structure of G - $\mathcal{O}_S$ -module, hence $\mathcal{L}ie(G/S)$ is a G - $\mathcal{O}_S$ -module (cf. I 4.7.1). If in addition G is *affine* over S , this amounts to saying, by I 4.7.2, that $\mathcal{L}ie(G/S)$ is an $\mathcal{A}(G)$ -comodule.

If T is a torus over S (IX Def. 1.3), one will say that T is *split* (“trivial” in the original) if it is isomorphic to $\mathbf{G}_{m,S}^r$ for some integer $r \geq 0$, and one will say that T is *trivialized* if one has fixed such an isomorphism or, more generally, an isomorphism $T \simeq \mathrm{D}_S(M)$, where M is a free abelian group of rank r .

Finally, recall (XII Def. 1.3) that a torus T of G is said to be *maximal* if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Theorem 2.5. *Let S be a scheme, G a group S -scheme affine and of finite presentation over S , with connected fibers, and s_0 a point of S . Suppose that G is flat over S at $\varepsilon(s_0)$ and that the geometric fiber $G_{\bar{s}_0}$ is (reduced and) reductive (resp. semisimple). Then there exist an open subset U of S containing s_0 , and a surjective étale morphism $S' \rightarrow U$, such that:*

- (i) G_U is smooth over S , with reductive (resp. semisimple) geometric fibers, of constant reductive rank and semisimple rank.
- (ii) $G_{S'}$ has a split maximal torus⁽²⁰⁾ T , and the Weyl group (cf. 6.3)

$$W_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / \underline{\mathrm{Centr}}_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / T$$

is finite over S' .

⁽²¹⁾Let $\pi : G \rightarrow S$ be the structural morphism and $\varepsilon : S \rightarrow G$ the unit section. Since G is flat over S at $\varepsilon(s_0)$ and $G_{\bar{s}_0}$ reduced (hence smooth over $\bar{s}_0$, cf. VI_A 1.3.1), G is smooth over S at the point $\varepsilon(s_0)$ (EGA IV₄, 17.5.1), i.e. there exists an open neighborhood V of $\varepsilon(s_0)$ such that $\pi|_V$ is smooth. Then $S' = \varepsilon^{-1}(V)$ is an open subset of S , and $G_{S'}$ is smooth over S' at the points of $\varepsilon(S')$. Since G has connected fibers, $G_{S'}$ is smooth over S' , by VI_B, 3.10. Thus, replacing S by S' , one may suppose that G is smooth over S .

By Exp. XI, th. 4.1, the functor of subgroups of multiplicative type of G is representable by an S -scheme $\mathcal{M}$, smooth and separated over S . Let r_0 be the rank of $G_{\bar{s}_0}$ and consider the open subscheme $\mathcal{M}_{r_0}$ of $\mathcal{M}$ which represents the functor of subtori of rank r_0 of G (IX 1.4). Smoothness implies that $\mathcal{M}_{r_0}$ has a rational point over a finite separable extension of $\kappa(s_0)$ (cf. EGA IV₄, 17.15.10 (iii)). Thus there exists an étale $S' \rightarrow S$ equipped with a point s'_0 mapping to s_0 such that $G_{s'_0}$ has a torus of rank r_0 . Therefore, replacing S by S' , one may suppose that G_{s_0} has a torus of rank r_0 .

By “Hensel’s lemma” (cf. XI, 1.10), the smoothness of $\mathcal{M}_{r_0}$ allows one to lift this torus to an S' -torus T of G , where $S' \rightarrow S$ is étale and equipped with a point s'_0 mapping to s_0 . By Exp. X, 4.5 (see also 6.1,⁽²²⁾), there exists an étale morphism $f : S'' \rightarrow S'$ splitting T and such that $f^{-1}(s'_0) \neq \emptyset$. Since an étale morphism is open and the assertions of (i) are local for the étale topology, one may therefore suppose that G has a split torus T ,⁽²³⁾ maximal at s_0 . Let us therefore write $T = \mathrm{D}_S(M)$ and let

$$\mathfrak{g} = \coprod_{m \in M} \mathfrak{g}^m$$

be the decomposition of $\mathfrak{g} = \mathcal{L}ie(G/S)$ under $\mathrm{Ad}(T)$ (Exp. I, 4.7.3).

⁽²⁰⁾**N.D.E.** “Split” has been added.

⁽²¹⁾**N.D.E.** The proof has been expanded.

⁽²²⁾**N.D.E.** Which is independent of the rest of this exposé.

⁽²³⁾**N.D.E.** Throughout this volume, “trivial torus” has been replaced by “split torus”.

One puts $\mathfrak{t} = \mathcal{L}ie(T)$ and, for every $m \in M$, denotes by $\mathfrak{g}^m(s_0) = \mathfrak{g}^m \otimes_{\mathcal{O}_S} \kappa(s_0)$.⁽²⁴⁾ Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}^m(s_0) \neq 0$.⁽²⁵⁾ Since $G_{\bar{s}_0}$ is reductive, one has $\mathfrak{g}^0(s_0) = \mathfrak{t}(s_0)$, hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha(s_0).$$

Since the modules in question are locally free, one may, restricting S if necessary, suppose that the $\mathfrak{g}^\alpha$ are free and

$$\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Recall, cf. Exp. XII, 1.12, that a group of multiplicative type has a unique maximal torus (this is, moreover, almost trivial by descent, the diagonalizable case being obvious). Let then T_α be the maximal torus of the kernel of α and $Z_\alpha = \underline{\text{Centr}}_G(T_\alpha)$.⁽²⁶⁾ By 2.2, Z_α is affine and smooth over S , with connected fibers, and by 1.12 its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3, there therefore exists an open subset U_α of S containing s_0 , such that $Z_\alpha|_{U_\alpha}$ has reductive fibers.

Put $U = \bigcap_{\alpha \in R} U_\alpha$. By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$, and the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ is identified with R , so that

$$\text{rgred}(G_{\bar{s}}) = \dim(T) = \text{rg}(M), \quad \text{rgss}(G_{\bar{s}}) = \text{rg}(R) \quad (\text{cf. 1.10}).$$

One has therefore proved (i) and the first assertion of (ii); it remains to prove that the Weyl group $W_{G_U}(T_U)$ is *finite* over U , i.e. “that it has the same number of points in each geometric fiber” (cf. SGA 1, I 10.9 and EGA IV₃, 15.5.1 and IV₄, 8.12.4).

For this, it suffices to note that the geometric fiber of this group at $s \in U$ is determined by the situation $R \subset M$, as the constant group associated with the “abstract Weyl group of this root system”, and in particular is independent of the point s , cf. *Bible*, § 11.3, th. 2 (see also Exp. XXII, no. 3).

Corollary 2.6. *Let G be an S -group affine and smooth over S , with connected fibers. The set of points $s \in G$ such that $G_{\bar{s}}$ is reductive (resp. semisimple) is an open subset U of S , and the functions*

$$s \longmapsto \text{rgred}(G_{\bar{s}}/\bar{s}), \quad s \longmapsto \text{rgss}(G_{\bar{s}}/\bar{s})$$

are locally constant on U .

Definition 2.7. An S -group (= group S -scheme) G is said to be *reductive* (resp. *semisimple*) if it is affine and smooth over S , with connected, reductive (resp. semisimple) geometric fibers.

The property of being reductive (resp. semisimple) for an S -group G is stable under base change and local for the (fpqc) topology.

2.8. — Let G be a reductive S -group. For every torus (resp. maximal torus) Q of G , $Z(Q) = \underline{\text{Centr}}_G(Q)$ is reductive (resp. is Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $\underline{\text{Centr}}_G(Q)$, one deduces that Q is contained (locally for the étale topology) in a maximal torus.

Remark 2.9. Using, as usual, the technique of EGA IV₃, § 8, the hypotheses of finite presentation, and theorem 2.5, one sees that if G is a reductive group over S , there exists an open covering $\{U_i\}$ of S such that each G_{U_i} comes by base change from a reductive group over a noetherian ring (in fact, an algebra of finite type over $\mathbf{Z}$). Similarly, if G has a split maximal torus T , one may in addition suppose that T_{U_i} comes from a split maximal torus of the preceding group, . . .

⁽²⁴⁾**N.D.E.** The preceding sentence has been added.

⁽²⁵⁾**N.D.E.** Note that, since $\mathfrak{g}$ is a finite locally free $\mathcal{O}_S$ -module, R is a finite set.

⁽²⁶⁾**N.D.E.** The following sentence has been expanded.

3. Roots and root systems of reductive group schemes

3.1. — Let S be a scheme, T an S -torus acting linearly on a locally free $\mathcal{O}_S$ -module $\mathcal{F}$ of finite type (cf. I, § 4.7). For every character α of T (i.e. $\alpha \in \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$), one defines a subfunctor of $W(\mathcal{F})$ by

$$W(\mathcal{F})^\alpha(S') = \{x \in W(\mathcal{F})(S') \mid tx = \alpha(t)x \text{ for every } t \in T(S''), S'' \longrightarrow S'\}.$$

Lemma. ⁽²⁷⁾ Then $W(\mathcal{F})^\alpha = W(\mathcal{F}^\alpha)$, where $\mathcal{F}^\alpha$ is a submodule of $\mathcal{F}$, locally a direct summand of $\mathcal{F}$, hence also locally free.

Indeed, the assertion is local for the (fpqc) topology, and one may suppose $T = D_S(M)$, where M is a finitely generated (free) abelian group. Then α is identified with a locally constant function from S to M (Exp. VIII 1.3), and, restricting S if necessary, one may suppose that this function is constant. One is then reduced to Exp. I, 4.7.3.

Definition 3.2. Let S be a scheme, G a group S -scheme smooth with connected fibers,⁽²⁸⁾ and T a torus of G . One puts $\mathfrak{g} = \text{Lie}(G/S)$ and makes T act on $\mathfrak{g}$ through the adjoint representation of G .

One says that the character α of T is a *root* of G with respect to T if the following equivalent conditions are satisfied:

- (i) For each $s \in S$, $\alpha_{\bar{s}}$ is a root of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ (1.10).
- (ii) α is nontrivial on each fiber and $\mathfrak{g}^\alpha(s) \neq 0$ for each $s \in S$.

Lemma 3.3. Let S be a scheme, T an S -torus, α a character of T . The following conditions are equivalent:

- (i) α is nontrivial on each fiber, that is: for every $s \in S$, $\alpha_{\bar{s}}$ is distinct from the unit character of $T_{\bar{s}}$.
- (ii) For every $S' \longrightarrow S$, $S' \neq \emptyset$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.
- (iii) The morphism α is faithfully flat.

⁽²⁹⁾It is clear that (ii) $\Rightarrow$ (i), and one easily sees that (iii) $\Rightarrow$ (i). One has (i) $\Rightarrow$ (ii), for if $s' \in S'$ lies above s and $\alpha_{s'}$ is the unit character, then so is α_s . Finally, to prove (i) $\Rightarrow$ (iii), one reduces to the case where T is diagonalizable and concludes by Exp. VIII 3.2 a).

3.4. — ⁽³⁰⁾Let G be a *reductive* group S -scheme, T a maximal torus of G . Let α be a root of G with respect to T . Then, by 2.5.0 and 1.12, $\mathfrak{g}^\alpha$ is a locally free $\mathcal{O}_S$ -module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T . In particular, if G has semisimple rank 1, by 1.11 and 1.12 one has:

Lemma 3.5. Let S be a scheme, G a reductive S -group of semisimple rank 1, T a maximal torus of G . If α is a root of G with respect to T , then $-\alpha$ is also one and one has

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank 1.

⁽²⁷⁾**N.D.E.** This statement has been made into a lemma, to emphasize it.

⁽²⁸⁾**N.D.E.** The original added the hypothesis that G be of finite presentation over S , which does not seem to be used in what follows. In any case, since G is smooth over S and has connected fibers, it is quasi-compact and separated over S (VI_B, 5.5), hence of finite presentation over S .

⁽²⁹⁾**N.D.E.** The original, which invoked Exp. IX, 5.2, has been simplified. Note, however, that loc. cit., 5.3 shows that if S is connected, the conditions of the lemma hold as soon as (i) holds at one point s of S .

⁽³⁰⁾**N.D.E.** The following has been modified, to recall the hypotheses of 3.2 and add that T is a maximal torus.

Definition 3.6. Let S be a scheme, G a reductive S -group, T a maximal torus of G , R a set of roots of G with respect to T . One says that R is a *root system* of G with respect to T if the following equivalent conditions are satisfied:

- (i) For each $s \in S$, $\alpha \mapsto \alpha_s$ is a bijection from R onto the set of roots of G_s with respect to T_s .
- (ii) The elements of R are distinct on each fiber (i.e. if $\alpha, \alpha' \in R$, $\alpha \neq \alpha'$, then $\alpha - \alpha'$ ($= \alpha\alpha'^{-1}$) is nontrivial on each fiber) and, for each $s \in S$, one has

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \text{Card}(R).$$

- (iii) One has $\mathfrak{g} = \mathfrak{t} \oplus \coprod_{\alpha \in R} \mathfrak{g}^\alpha$.

The equivalence of these various conditions is trivial.

Lemma 3.7. *Let S be a scheme, G a reductive S -group, T a maximal torus of G , R a root system of G with respect to T . Every root of G with respect to T is locally on S equal to an element of R .*

This is apparent from condition (iii).

Put $\mathcal{M} = \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$; this is a twisted constant group S -scheme (Exp. X 5.6). If G has a root system R with respect to T , the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism $R_S \rightarrow \mathcal{M}$, where R_S is the constant S -scheme defined by R ; thanks to condition (ii), one easily sees that this morphism is an open and closed immersion whose image is precisely $\bigcup_{\alpha \in R} \alpha(S)$ (each $\alpha \in R$ being considered as a section $S \rightarrow \mathcal{M}$).

Let $\mathcal{R}$ be the *functor of roots* of G with respect to T : by definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$ for every $S' \rightarrow S$; if $S' = \emptyset$, one puts $\mathcal{R}(\emptyset) = \{e\}$, and if $S' \neq \emptyset$, the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$, and hence, by 3.7, one has

$$\mathcal{R}(S') = \text{Hom}_{\text{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is representable by R_S .

If now one no longer necessarily supposes that G has a root system relative to T , $\mathcal{R}$ is in any case a subsheaf of $\mathcal{M}$ for (fpqc). Locally for this topology, G has a root system with respect to T (take, for example, T split and use the proof of Th. 2.5). By Exp. IV 4.6.8 and the theory of descent of open (resp. closed) subschemes,⁽³¹⁾ one obtains:

Proposition 3.8. *Let S be a scheme, G an S -group, T a maximal torus of G . The functor $\mathcal{R}$ of roots of G with respect to T is representable by a finite twisted constant S -scheme (Exp. X 5.1) which is an open and closed subscheme of $\underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$.*

For $R \subset \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ to be a root system of G with respect to T , it is necessary and sufficient that the morphism $R_S \rightarrow \underline{\text{Hom}}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ defined by the preceding inclusion induce an isomorphism $R_S \xrightarrow{\sim} \mathcal{R}$.

3.9. — Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T (i.e. a section of $\mathcal{R}$). Consider the kernel $\text{Ker}(\alpha)$ of R , its unique maximal torus T_α , and the centralizer of the latter, $Z_\alpha = \underline{\text{Centr}}_G(T_\alpha)$. This is an S -group closed in G , reductive (2.8), of *semisimple rank 1* (1.12). Moreover,

$$\mathcal{L}ie(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

hence $\{\alpha, -\alpha\}$ is a root system of Z_α with respect to T .

3.10. — Let S be a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T . If $q \in \mathbf{Q}$ and $q\alpha$ is a root of G with respect to T , then $q = 1$ or $q = -1$. This follows at once from 1.12.

4. Roots and vector group schemes

⁽³¹⁾N.D.E. See SGA 1, VIII 4.4 or EGA IV₂, 2.7.1.

4.1. — Let S be a scheme, $\mathcal{F}$ a *locally free* $\mathcal{O}_S$ -module of finite type. The S -scheme $W(\mathcal{F})$ is *smooth* over S . Its Lie algebra is canonically isomorphic to $\mathcal{F}$. Indeed, one has a canonical isomorphism

$$W(\mathcal{F}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(W(\mathcal{F})/S) = W(\mathrm{Lie}(W(\mathcal{F})/S)).$$

(Exp. II, 4.4.1 and 4.4.2). We shall always identify $\mathcal{F}$ and $\mathrm{Lie}(W(\mathcal{F})/S)$.

Lemma 4.2. *Let S be a scheme, V a vector bundle over S , smooth over S . There then exists a unique isomorphism of $\mathcal{O}_S$ -modules*

$$\exp : W(\mathrm{Lie}(V/S)) \xrightarrow{\sim} V$$

inducing the identity on Lie algebras.⁽³²⁾

Indeed, one has $V = V(\mathcal{F})$ for some quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Since V is smooth over S , $\mathcal{F} \simeq \omega_{V/S}^1$ is locally free of finite type, and hence

$$\underline{\mathrm{Lie}}(V/S) = V(\omega_{V/S}^1) \simeq W(\mathrm{Lie}(V/S)).$$

Moreover, by Exp. II loc. cit., one has a canonical isomorphism

$$V \xrightarrow{\sim} \underline{\mathrm{Lie}}(V/S) \simeq W(\mathrm{Lie}(V/S)),$$

and the uniqueness of $\exp$ follows at once, since W is fully faithful.

4.3. — Notations. If V is a vector bundle over S , $V^\times$ will denote the open subset of V obtained by removing the zero section. Let us write the group law of V in multiplicative notation. The action of $\mathbf{O}_S$ on V defining the module structure will then be written exponentially:

$$\mathbf{O}_S \times_S V \longrightarrow V, \quad (x, v) \longmapsto v^x.$$

One has $(vv')^x = v^x v'^x$, $v^{x+x'} = v^x v^{x'}$, $v^{xx'} = (v^x)^{x'}$. In particular, if one restricts the operator law to $\mathbf{G}_{m,S}$, $V^\times$ is stable and is therefore equipped with the structure of an object with operator group $\mathbf{G}_{m,S}$. One will again denote this law by $(z, v) \longmapsto v^z$.

Definition 4.4.1. ⁽³³⁾Let $\mathcal{L}$ be an invertible module over S and $W(\mathcal{L})$ the associated vector bundle. Then $W(\mathcal{L})^\times$ is a (locally trivial) principal homogeneous bundle under $\mathbf{G}_{m,S}$. One puts $\Gamma(S, \mathcal{L})^\times = W(\mathcal{L})^\times(S)$.

⁽³²⁾**N.D.E.** The proof of the original has been retained; one may also expand it as follows. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module, $V = V(\mathcal{F})$. Let π denote the projection $V \longrightarrow S$ and ε the zero section $S \longrightarrow V$. Then $\Omega_{V/S}^1 = \pi^* \mathcal{F}$, whence

$$(1) \quad \omega_{V/S}^1 = \varepsilon^* \Omega_{V/S}^1 \simeq \varepsilon^* \pi^* \mathcal{F} \simeq \mathcal{F},$$

and hence $\mathrm{Lie}(V/S) = (\omega_{V/S}^1)^\vee \simeq \mathcal{F}^\vee$. If one supposes V smooth over S , then, by (1), $\mathcal{F}$ is locally free of finite type, and hence

$$(2) \quad V = V(\mathcal{F}) \simeq W(\mathcal{F}^\vee) \simeq W(\mathrm{Lie}(V/S)).$$

Now, if U is an S -scheme equipped with a section $\tau : S \longrightarrow U$, and one has given an isomorphism $\phi : V \xrightarrow{\sim} U$ of “pointed” S -schemes, i.e. such that $\phi \circ \varepsilon = \tau$ (for example if U is an S -group), then ϕ induces an isomorphism $d\phi : \mathcal{F}^\vee \xrightarrow{\sim} \mathrm{Lie}(U/S)$, whence an isomorphism ψ :

$$\begin{array}{ccc} W(\mathcal{F}^\vee) & \xlongequal{\quad} & V(\mathcal{F}) \xrightarrow{\phi} U \\ & \nwarrow \simeq & \nearrow \simeq \\ & W(d\phi^{-1}) & W(\mathrm{Lie}(U/S)) \end{array}$$

which allows one to identify U with $W(\mathrm{Lie}(U/S))$. Since the functor W is fully faithful, there exists a unique isomorphism $\exp : W(\mathrm{Lie}(U/S)) \xrightarrow{\sim} U$ such that $\mathrm{Lie}(\psi^{-1} \circ \exp) = \mathrm{id}$. In fact, one can see directly that $\mathrm{Lie}(\psi) = \mathrm{id}$, so that $\exp = \psi$.

⁽³³⁾**N.D.E.** For subsequent references, 4.4 has been changed into 4.4.1 and 4.4.2.

Scholium 4.4.2. *There is a bijective correspondence between isomorphisms of $\mathcal{O}_S$ -modules $\mathcal{O}_S \simeq \mathcal{L}$, isomorphisms of $\mathbf{O}_S$ -modules $\mathbf{O}_S \simeq \mathbf{W}(\mathcal{L})$,⁽³⁴⁾ and sections $S \rightarrow \mathbf{W}(\mathcal{L})^\times$.*

This correspondence is given by $f \mapsto \mathbf{W}(f) \mapsto \mathbf{W}(f)(1)$. It is compatible with extension of the base. One may therefore regard $\mathbf{W}(\mathcal{L})^\times$ as the “scheme of trivializations of $\mathbf{W}(\mathcal{L})$ ”.

4.5. — Let S be a scheme, T a torus over S , P an S -group with operator group $\mathbf{G}_{\mathbf{m},S}$ (for example, a vector bundle), α a character of T . One denotes by $T \cdot_\alpha P$ the semidirect product of P by T , where T acts on P by the composite morphism

$$T \xrightarrow{\alpha} \mathbf{G}_{\mathbf{m},S} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(P).$$

Definition 4.6. Let S be a scheme, G a group S -scheme, T a subgroup of G , α a character of T , $\mathcal{L}$ an $\mathcal{O}_S$ -module. Let

$$p : \mathbf{W}(\mathcal{L}) \rightarrow G$$

⁽³⁵⁾be a morphism of group S -functors. One says that p is *normalized by T with multiplier α* if it satisfies the following equivalent conditions:

- (i) p is a morphism of objects with operator group T , if one makes T act on $\mathbf{W}(\mathcal{L})$ by α and on G by inner automorphisms. In other words, for every $S' \rightarrow S$ and all $t \in T(S')$ and $x \in \mathbf{W}(\mathcal{L})(S') = \Gamma(S', \mathcal{L} \otimes \mathcal{O}_{S'})$, one has

$$\mathrm{int}(t)p(x) = p(\alpha(t)x).$$

- (ii) The morphism $T \cdot_\alpha \mathbf{W}(\mathcal{L}) \rightarrow G$ defined by the product in G (i.e. by $(t, x) \mapsto t \cdot p(x)$) is a morphism of groups.

Lemma 4.7. *Under the conditions of 4.6, suppose in addition that G is smooth and has connected fibers,⁽³⁶⁾ T is a maximal torus of G , and $\mathcal{L}$ invertible. If p is a monomorphism and α nonzero on each fiber, then α is a root of G with respect to T .*

Indeed, $\mathrm{Lie}(p) : \mathcal{L} \rightarrow \mathfrak{g}$ is a monomorphism which factors through $\mathfrak{g}^\alpha$.

Proposition 4.8. *Under the conditions of 4.7, suppose that G is reductive and that p is a monomorphism. Then α is a root of G with respect to T , and $\mathrm{Lie}(p)$ induces an isomorphism*

$$\mathrm{Lie}(p) : \mathcal{L} \xrightarrow{\sim} \mathfrak{g}^\alpha.$$

Indeed, by 4.7 and the fact that $\mathfrak{g}^\alpha$ is invertible, it suffices to prove that α is nonzero on each fiber. Let therefore $s \in S$ be such that $\alpha_{\bar{s}} = 0$ ($= 1$ in multiplicative notation). If X is a nonzero section of $\mathcal{L} \otimes_S \kappa(\bar{s})$, $p(X)$ is a unipotent element $\neq e$ of $G(\bar{s})$ which centralizes $T_{\bar{s}}$, which is impossible, since the latter is its own centralizer.

Corollary 4.9. *Under the conditions of 4.8, there exists a monomorphism of groups with operators T (i.e. normalized by T with multiplier α)*

$$\mathbf{W}(\mathfrak{g}^\alpha) \rightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^\alpha \rightarrow \mathfrak{g}$.

We shall soon see that 4.9 in fact holds whenever G is a reductive group and α a root of G with respect to T , and that such a morphism is unique.

⁽³⁴⁾**N.D.E.** Here the vector bundle $\mathbf{W}(\mathcal{L})$ is identified with the functor in $\mathbf{O}_S$ -modules which it represents.

⁽³⁵⁾**N.D.E.** Here $\mathbf{W}(\mathcal{L})$ has been written (with W in bold), since, for an arbitrary $\mathcal{O}_S$ -module $\mathcal{L}$, $\mathbf{W}(\mathcal{L})$ is not necessarily representable. But in what follows, $\mathcal{L}$ will be supposed locally free of finite rank (and even invertible), in which case the functor is representable by the vector bundle $\mathbf{V}(\mathcal{L}^\vee)$, and it will simply be denoted $\mathbf{W}(\mathcal{L})$.

⁽³⁶⁾**N.D.E.** Cf. N.D.E. (28).

Reminder 4.10. Let k be an algebraically closed field, G a reductive k -group, T a maximal torus of G , α a root of G with respect to T . There exists a monomorphism

$$p : \mathbf{G}_{a,k} \longrightarrow G$$

normalized by T with multiplier α .

See *Bible*, § 13.1, th. 1.

4.11. — Let us finish this section with a technical result which will be useful to us. Let S be a scheme and $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module. Let q be an integer > 0 such that $x \mapsto x^q$ is an endomorphism of the S -group $\mathbf{G}_{a,S}$. (If $S \neq \emptyset$, one has $q = 1$, or $q = p^n$, p being a prime number zero on S ; this follows at once from the following elementary fact: the gcd of the binomial coefficients $\binom{q}{i}$, for $i \neq 0, q$, is p if $q = p^n$, p prime, and 1 otherwise.) The morphism defined by the q th power

$$\mathcal{L} \longrightarrow \mathcal{L}^{\otimes q}$$

is a morphism of sheaves of abelian groups. By base change, it defines a morphism of group S -schemes:

$$W(\mathcal{L}) \longrightarrow W(\mathcal{L}^{\otimes q}).$$

In particular, if $\mathcal{L}'$ is another invertible module and one has a morphism of $\mathcal{O}_S$ -modules

$$h : \mathcal{L}^{\otimes q} \longrightarrow \mathcal{L}',$$

one deduces a morphism of group S -schemes:

$$W(\mathcal{L}) \longrightarrow W(\mathcal{L}'), \quad x \mapsto h(x^q).$$

With these notations, one has

Proposition 4.12. *Let S be a scheme, T (resp. T') an S -torus, $\mathcal{L}$ (resp. $\mathcal{L}'$) an invertible $\mathcal{O}_S$ -module, α (resp. α') a character of T (resp. T').⁽³⁷⁾ Let $f : T \longrightarrow T'$ be a morphism of groups and $g : W(\mathcal{L}) \longrightarrow W(\mathcal{L}')$ a morphism of S -schemes (not necessarily a morphism of S -groups) satisfying the following condition:*

$$(\star) \quad g(\alpha(t)x) = \alpha'(f(t))g(x)$$

for all $x \in W(\mathcal{L})(S')$, $t \in T(S')$, $S' \longrightarrow S$. Let $s_0 \in S$ be such that $\alpha_{\bar{s}_0} \neq 0$.

- a) Suppose that g sends the zero section to the zero section and that for every integer $n > 0$ one has $(\alpha' \circ f)_{\bar{s}_0} \neq n\alpha_{\bar{s}_0}$. Then $g = 0$ in a neighborhood of s_0 .
- b) Suppose that g is a morphism of groups such that $g_{\bar{s}_0} \neq 0$. There then exist an open subset U of S containing s_0 and an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,U}$ and $(\alpha' \circ f)_U = q\alpha_U$.
- c) Suppose that $(\alpha' \circ f)_{\bar{s}_0} = q\alpha_{\bar{s}_0}$, where q is an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$. There then exist an open subset U of S containing s_0 and a unique morphism of $\mathcal{O}_S$ -modules

$$h : \mathcal{L}^{\otimes q}|_U \longrightarrow \mathcal{L}'|_U$$

such that g_U is the composite morphism

$$W(\mathcal{L})_U \xrightarrow{x \mapsto x^q} W(\mathcal{L}^{\otimes q})_U \xrightarrow{W(h)} (\mathcal{L}')_U.$$

Let us prove (a). Since the conclusion is local on S , one may suppose that $W(\mathcal{L}) = W(\mathcal{L}') = \mathbf{G}_{a,S}$ and hence that g is expressed as a polynomial

$$g(X) = \sum_{n \geq 0} a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

⁽³⁷⁾**N.D.E.** “Consider the semidirect product $T \cdot_{\alpha} W(\mathcal{L})$, resp. $T' \cdot_{\alpha'} W(\mathcal{L}')$ ” has been deleted. On the other hand, the hypothesis in (b) that g be a morphism of groups, combined with $(\star)$, amounts to saying that the morphism $(t, x) \mapsto (f(t), g(x))$ is a morphism of groups from $T \cdot_{\alpha} W(\mathcal{L})$ to $T' \cdot_{\alpha'} W(\mathcal{L}')$.

Condition $(\star)$, which relates f and g , is written as an identity in $\Gamma(S', \mathcal{O}_{S'})[X]$:

$$\sum_{n \geq 0} a_n \alpha'(f(t)) X^n = \sum_{n \geq 0} a_n \alpha(t)^n X^n,$$

i.e. for every $n \geq 0$, every $S' \rightarrow S$ and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

For each $n \geq 0$, let S_n be the set of $s \in S$ such that $(\alpha' \circ f)_{\bar{s}} = n\alpha_{\bar{s}}$. One knows (Exp. IX 5.3) that the S_n are open and closed, and that $(\alpha' \circ f)_{S_n} = n\alpha_{S_n}$. Moreover, since $\alpha_{\bar{s}_0} \neq 0$, one may, restricting S if necessary, suppose that α is nonzero on each fiber (same reference), which implies that the S_n are disjoint. Restricting S if necessary, one may therefore suppose that one is in one of the following two cases: there exists an n such that $S = S_n$, or all the S_n are empty.

Let $m \geq 0$ be such that $S_m = \emptyset$; I say that then $a_m = 0$; indeed, $\alpha' \circ f$ and $m\alpha$ are distinct on each fiber of S , and one has:

Lemma 4.13. *Let S be a scheme, T an S -torus, α and α' two characters of T distinct on each fiber; there exists a family $\{S_i \rightarrow S\}$ covering for (fpqc), and, for each i , a $t_i \in T(S_i)$ such that $\alpha(t_i) - \alpha'(t_i) = 1$.*

The lemma is trivial, by reduction to the diagonalizable case, then to the case $T = \mathbf{G}_{m,S}$.

Let us resume the proof of proposition 4.12; we have just proved (a). In cases (b) and (c), there exists an n such that $S = S_n$ ($n = q$ in (c)). By the preceding result, one therefore has $a_m = 0$ for $m \neq n$, which proves that g is written

$$g(X) = a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

This proves (c) at once. In case (b), one knows that $a_n(s_0) \neq 0$, and one may therefore suppose a_n invertible on S , which implies that $x \mapsto x^n$ is an endomorphism of $\mathbf{G}_{a,S}$ (by the hypothesis of (b)) and completes the proof.

5. An instructive example

5.1. — Let k be an algebraically closed field of characteristic 0. Put $A = k[t]$, the polynomial ring in one variable over k , and $S = \text{Spec}(A)$. Consider the following Lie algebra $\mathfrak{g}$ over $\mathcal{O}_S$: as a module, it is free of dimension 3, with basis $\{X, Y, H\}$; its multiplication table is

$$[X, Y] = 2tH, \quad [H, X] = X \quad \text{and} \quad [H, Y] = -Y.$$

For $s \in S$, $s \neq s_0$ (the point defined by $t = 0$), the fiber $\mathfrak{g}(s) = \mathfrak{g} \otimes_A \kappa(s)$ is isomorphic to the Lie algebra of the group $\text{PGL}_{2, \kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2. — Let G_1 be the group scheme of automorphisms of $\mathfrak{g}$: for every $S' \rightarrow S$, $G_1(S')$ is the group of automorphisms of the Lie $\mathcal{O}_{S'}$ -algebra $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. This is a closed subscheme of the group $\text{GL}(\mathfrak{g})$ of automorphisms of the $\mathcal{O}_S$ -module $\mathfrak{g}$. Let $S' \rightarrow S$ and $u \in \text{M}_3(\Gamma(S', \mathcal{O}_{S'}))$, considered as an endomorphism of the $\mathcal{O}_S$ -module $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$:

$$\begin{aligned} u(X) &= aX + bY + eH, \\ u(Y) &= b'X + a'Y + e'H, \\ u(H) &= cX + c'Y + dH. \end{aligned}$$

One sees at once that u is a section of G_1 if and only if $\det(u)$ is invertible and one has the relations:⁽³⁸⁾

$$\begin{aligned} (1) \quad a(d-1) &= ec, & (1') \quad a'(d-1) &= e'c', \\ (2) \quad b(d+1) &= ec', & (2') \quad b'(d+1) &= e'c, \\ (3) \quad e &= 2t(bc - ac'), & (3') \quad e' &= 2t(b'c' - a'c), \\ (4) \quad 2tc &= eb' - ae', & (4') \quad 2tc' &= be' - ea', \\ (5) \quad 2t(aa' - bb') &= 2td. \end{aligned}$$

Lemma 5.3. *Relations (1), (1'), (2), (2') imply*

$$\det(u) = aa'(2-d) + bb'(d+2), \quad aa' - bb' = d \cdot \det(u).$$

Indeed, the first assertion follows at once by substituting relations (1), (1'), (2), (2') into the expansion of $\det(u)$:

$$\begin{aligned} \det(u) &= aa'd + be'c + b'c'e - a'ec - ae'c' - bb'd \\ &= aa'd + bb'(d+1) + bb'(d+1) - aa'(d-1) - bb'd - aa'(d-1) \\ &= aa'(d-d+1-d+1) + bb'(d+1+d+1-d) \\ &= aa'(2-d) + bb'(d+2). \end{aligned}$$

Multiplying this relation by d , one obtains

$$d \cdot \det(u) = aa'(2d-d^2) + bb'(d^2+2d).$$

But the relation $(1) \times (1') = (2) \times (2')$ gives at once

$$aa'(d-1)^2 = bb'(d+1)^2.$$

Combining the preceding two relations, one finds at once the second desired formula.

5.4. — Consider then $G = G_1 \cap \mathrm{SL}(\mathfrak{g})$. This is the closed subgroup of G_1 defined by the equation $\det(u) = 1$. It is therefore an affine group over S .

Proposition 5.5. *The group G is smooth over S .*

To prove the proposition, one will need the following lemmas.

Lemma 5.6. *Let U be the open subset of $\underline{\mathrm{End}}_A(\mathfrak{g}) \simeq \mathrm{W}(\mathrm{M}_3(\mathcal{O}_S))$ defined by the condition “the product ad is invertible”, i.e. the open subset $\underline{\mathrm{End}}_A(\mathfrak{g})_f$, where f is the function defined by $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations:*

$$(1), \quad (2), \quad (2'), \quad (3), \quad (3') \quad \text{and} \quad (D) : aa' - bb' = d.$$

First, it is clear that these six relations are satisfied by every “point” of G (lemma 5.3), in particular by every “point” of $U \cap G$. Conversely, one must show that if $u \in U(S')$ (for every $S' \rightarrow S$), and u satisfies the six conditions of the statement, then $\det(u) = 1$ and u also satisfies (1'), (4), (4') and (5).

First, one has $(D) \Rightarrow (5)$. By (2) and (2'), one has

$$bb'(d+1) = bce' = b'c'e.$$

But by (3) and (3'), writing $2t(bc - ac')(b'c' - a'c)$ in two ways, one has

$$(bc - ac')e' = (b'c' - a'c)e.$$

Combining this with the preceding relation gives $ac'e' = a'ce$. But by (1), $a'ce = a'a(d-1)$, which proves $a(a'(d-1) - e'c') = 0$ and implies (1'), since a is assumed invertible.

⁽³⁸⁾**N.D.E.** The equality $[u(X), u(Y)] = 2tu(H)$ (resp. $[u(H), u(X)] = u(X)$, resp. $[u(H), u(Y)] = -u(Y)$) gives relations (4), (4') and (5) (resp. (1–3), resp. (1'–3')).

Thus (1), (2), (2') and (1') hold, hence by lemma 5.3 and (D) one has $d(\det(u) - 1) = 0$. Since d is assumed invertible, this implies $\det(u) = 1$.

Let us prove (4) and (4'). Let us do so, for example, for (4'), the other calculation being deduced from it by symmetry. By (3), (3') and (D), one has at once

$$a'e + be' = -2t(aa' - bb')c' = -2tdc'.$$

Combining (1') and (2), one also has⁽³⁹⁾

$$a'e + be' = -d(be' - ea'),$$

which finishes the proof of (4'), since d is assumed invertible.

Lemma 5.7. *G is smooth over S along the unit section.*

By 5.6 and SGA 1, II 4.10, it suffices to prove that the differentials of the functions

$$\begin{aligned} a(d-1) - ec, & \quad b(d+1) - ec', & \quad b'(d+1) - e'c, \\ e - 2t(bc - ac'), & \quad e' - 2t(b'c' - a'c), & \quad aa' - bb' - d, \end{aligned}$$

at the points of the unit section of G are linearly independent. Now, denoting by a capital letter the differential of the corresponding lower-case letter, these are⁽⁴⁰⁾

$$D, \quad 2B, \quad 2B', \quad E + 2tC', \quad E' + 2tC, \quad A + A' - D,$$

which are indeed linearly independent modulo every $(t - \lambda)$, $\lambda \in k$.⁽⁴¹⁾

Lemma 5.8. *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

⁽⁴²⁾Indeed, since $s \neq s_0$, $\mathfrak{g}(s)$ is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$, and, on the other hand, one has $G_s = (G_1)_s$; now it is known that the group of automorphisms of the Lie algebra of PGL_2 over a field of characteristic 0 is PGL_2 itself, which is connected and semisimple.

Lemma 5.9. *The fiber G_{s_0} is solvable and has two connected components of the following form:*

$$G_{s_0}^0 = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \right\} \quad \text{and} \quad G_{s_0}^- = \left\{ \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix} \right\}.$$

Indeed, one has $e = e' = 0$, since $t = 0$ at s_0 . One then immediately solves equations (1), (1'), ..., (5) and (D).

Lemma 5.10. *$w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ is a section of G over S such that $w(s_0) \in G_{s_0}^-$.*

Let us now prove 5.5.⁽⁴³⁾ Denote by G^0 the union of the identity components of the fibers of G (that is, the complement of $G_{s_0}^-$); since G is smooth over S along the unit section (5.7), G^0 is an open subgroup of G , smooth over S , by VI_B, 3.10. Since, by translation, G is clearly smooth at the points of $w(S)$, G is smooth over S .

⁽³⁹⁾**N.D.E.** The sign has been corrected.

⁽⁴⁰⁾**N.D.E.** $A + A' + B$ has been corrected to $A + A' - D$.

⁽⁴¹⁾**N.D.E.** “ $(t - a)$, $a \in A$ ” has been corrected to “ $(t - \lambda)$, $\lambda \in k$ ”.

⁽⁴²⁾**N.D.E.** The following sentence has been slightly modified.

⁽⁴³⁾**N.D.E.** The following sentence has been modified, adding the reference to VI_B, 3.10.

5.11. — Consider the morphism $\mathbf{G}_{m,S} \rightarrow G^0$ defined by $z \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & 1/z & 0 \\ 0 & 0 & 1 \end{pmatrix}$. It is a monomorphism defining a torus T of G^0 . I say that one has

$$T = \underline{\text{Centr}}_G(T) = \underline{\text{Centr}}_{G^0}(T).$$

Indeed, it suffices to check the first equality. Since these are subgroups of G smooth over S , it suffices to check that they have the same geometric points. For the fibers at points $s \neq s_0$, this follows from the fact that $\text{PGL}_{2,\kappa(s)}$ is reductive and that T_s is a maximal torus of it for dimension reasons (cf. 1.11). On the fiber at s_0 , the calculation is immediate. In particular, it follows that T is a maximal torus of G and of G^0 .

5.12. — The section w of G defined in 5.9 normalizes T . It follows at once (cf. 2.4) that the Weyl group of G is isomorphic to $(\mathbf{Z}/2\mathbf{Z})_S$, and in particular finite over S .

$$W_G(T) = \underline{\text{Norm}}_G(T)/T = (\mathbf{Z}/2\mathbf{Z})_S.$$

On the other hand, $W_{G^0}(T)$ is *not finite* over S : it “lacks a point” above s_0 .

5.13. — The open immersion $G^0 \rightarrow G$ is not a closed immersion (since G^0 is dense in G); it is nevertheless an affine morphism (and hence G^0 is affine over S). Indeed, since $G_{s_0}^0$ is closed in G_{s_0} , which is closed in G , the complement U of $G_{s_0}^0$ in G is open; G^0 and U form an open covering of G , and it suffices to check that the immersions $G^0 \rightarrow G^0$ and $G^0 \cap U \rightarrow U$ are affine; for the first this is trivial, for the second one notes that $U \cap G^0$ is defined in U by the equation $b \neq 0$.

One has therefore constructed a smooth affine S -group G^0 , with connected fibers, having a maximal torus T which is its own centralizer and whose Weyl group $W_{G^0}(T)$ is *not finite* (compare with theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the course of the proof of 2.5, we used a result of Exp. XI on the local existence, for the étale topology, of maximal tori; the proof of Exp. XI uses a delicate representability result (XI 4.1). In the particular case which concerns us, one can give another proof, based on the ideas of Exp. XII no. 7, and of a much more elementary nature.

Proposition 6.1. *Let S be a scheme, G a group S -scheme smooth, affine and with connected fibers over S , and s_0 a point of S such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are their own centralizer. There exists an étale morphism $S' \rightarrow S$ covering s_0 , and a split maximal torus T of $G_{S'}$.*

First, one may suppose S affine.⁽⁴⁴⁾ Since G is of finite presentation over S , one may suppose S noetherian, then local, then henselian with separably closed residue field (cf. EGA IV, 8.12, § 8.8, and § 18.8). Put therefore $S = \text{Spec}(A)$, A henselian with separably closed residue field $k = \kappa(s_0)$. Choose a maximal torus T_0 of $G_0 (= G_k)$ (one exists, for example, because the scheme of maximal tori of G_0 is smooth over k , Exp. XII, 7.1 c)); since k is separably closed, T_0 is split (cf. X 1.4), and is therefore given by a monomorphism of groups

$$f_0 : \mathbf{G}_{m,k}^r \rightarrow G_0.$$

Let m be an integer > 1 prime to the characteristic of k . By Exp. VIII 6.7, for every $h > 0$, $\underline{\text{Centr}}_{G_0}(m^h T_0)$ is representable by a closed subscheme of G_0 . Since the $m^h T_0$ are schematically dense in T_0 (cf. Exp. IX 4.10) and G_0 is noetherian, there exists an h such that

$$\underline{\text{Centr}}_{G_0}(m^h T_0) = \underline{\text{Centr}}_{G_0}(T_0) = T_0.$$

⁽⁴⁴⁾**N.D.E.** The preceding sentence has been added, as well as the reference to EGA IV in what follows.

Put $n = m^h$; since n is invertible on S , ${}_n\mathbf{G}_{m,S}$ is isomorphic to $(\mathbf{Z}/n\mathbf{Z})_S$; f_0 therefore defines a monomorphism of groups

$$u_0 : (\mathbf{Z}/n\mathbf{Z})_k^r \longrightarrow G_0$$

such that $\underline{\text{Centr}}_G(u_0) = T_0$. Now the S -functor

$$P = \underline{\text{Hom}}_{S\text{-gr.}}((\mathbf{Z}/n\mathbf{Z})_S^r, G)$$

is representable by an S -scheme of finite type (as a closed subscheme of r copies of ${}_nG = \underline{\text{Hom}}_{S\text{-gr.}}((\mathbf{Z}/n\mathbf{Z})_S, G) = \text{Ker}(G \xrightarrow{n} G)$). But P is smooth over S (Exp. IX 3.6), hence $u_0 \in P(k)$ lifts to a section $u \in P(S)$ (Hensel's lemma, Exp. XI 1.11):

$$u : (\mathbf{Z}/n\mathbf{Z})_S^r \longrightarrow G.$$

Consider $H = \underline{\text{Centr}}_G(u)$; it is a closed subgroup scheme of G , by Exp. VIII 6.5 e), and one has $H_0 = T_0$ by hypothesis.⁽⁴⁵⁾ Moreover, H is smooth over S : indeed, let $S' = \text{Spec}(A)$ be an affine scheme over S , $u' : ({}_n\mathbf{G}_{m,S'})^r \longrightarrow G_{S'}$ the morphism deduced from u by base change, $S'_J = \text{Spec}(A/J)$, where J is a square-zero ideal, and let $x \in H(S'_J)$; since G is smooth, x lifts to an element g of $G(S')$; then $v = \text{int}(g)(u')$ satisfies $v_J = u'_J$, and hence, by IX 3.2, there exists an element g' of $G(S')$ such that $g'_J = e$ and $\text{int}(g')(v) = u'$; then $h = g'g$ belongs to $H(S')$ and satisfies $h_J = x$.

Let then H^0 be the identity component of H ; it is a subgroup scheme of G , smooth with connected fibers, whose special fiber is a torus. By Exp. X 8.1, it is a torus, necessarily split (Exp. X 4.6). Put $H^0 = T$, and let $C = \underline{\text{Centr}}_G(T)$, which is a closed subgroup of G (Exp. VIII 6.5 e)), smooth (Exp. XI 2.4). Consider C^0 (in fact one has $C^0 = C$, but we do not need to know this); then $C^0 \supset T$, and these are two smooth groups with connected fibers. They coincide at s_0 , hence in a neighborhood. Restricting S if necessary, one may therefore suppose $C^0 = T$, hence a fortiori T maximal.

Remark 6.2. The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.

Proposition 6.3. *Let S be a scheme, G a group S -scheme smooth and of finite presentation over S , Q a subtorus of G .*

- (i) $\underline{\text{Centr}}_G(Q)$ and $\underline{\text{Norm}}_G(Q)$ are representable by closed subgroup schemes, smooth (and hence of finite presentation) over S .
- (ii) $\underline{\text{Centr}}_G(Q)$ is an open and closed subscheme of $\underline{\text{Norm}}_G(Q)$. The quotient $W_G(Q) = \underline{\text{Norm}}_G(Q)/\underline{\text{Centr}}_G(Q)$ is representable by an open subgroup scheme of $\underline{\text{Aut}}_{S\text{-gr.}}(Q)$, and is therefore a group S -scheme quasi-finite, étale and separated over S .
- (iii) For every $s \in S$, put

$$w(s) = \left| \underline{\text{Norm}}_{G(\bar{s})}(Q(\bar{s}))/\underline{\text{Centr}}_{G(\bar{s})}(Q(\bar{s})) \right|.$$

Then $s \mapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s .

By Exp. XI 6.11, $\underline{\text{Centr}}_G(Q)$ and $\underline{\text{Norm}}_G(Q)$ are representable by closed subschemes of finite presentation of G . These are smooth by Exp. XI 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exp. XI 5.9 and 5.10, whose proof in fact uses only (i), and not the delicate theorems Exp. XI 4.1 and 4.2.

⁽⁴⁵⁾**N.D.E.** The reference VIII 6.5 e) has been added above, and the following sentence has been expanded.

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